

CompareMLModelsV2:

a. Based on accuracy, neural network is the best one, with an accuracy of 0.98.

b.

-Naïve Bayesian: limited to some assumptions on distribution and independency.

-Linear regression: not suitable for fitting non-linear data.

-Polynomial of degree 2/3: features might not be in the form of 2(or 3)-degree polynomial relation with label. While neural network does not necessarily assume any data distribution form.

-KNN Classifier: good at handling low-dimension data, but might not accurate enough as neural network to model complicated input-output mappings.

- LDA/QDA: not so far away from the best one. Neural network performs better probably because it fits non-linear boundary better than LDA/QDA.

dbn.py:

a. Yes the code is using 5-fold cross validation

b. A 0.8-0.2 train-test split is used, so 20% of the dataset was used for training.

c. $150 * 0.2 = 30$

d. $150 * 0.8 = 120$

e. 4 features

f. 4 features

g. 3 classes

h. ['setosa', 'versicolor', 'virginica']